

Problem 15.8

A grandmother has a granddaughter entering university next year. Her granddaughter expects to remain in school for ten years and receive a PhD. This grandmother wishes to provide \$1,000 a year to her granddaughter for entertainment expenses. Assuming a 3.5% effective annual interest rate, how much does the grandmother have to deposit today to provide ten annual payments starting one year from now and continuing for ten years?

Solution

The payments begin one year from now and continue annually for ten years. Thus, this is an annuity-immediate with payment amount 1000, interest rate

$$i = 0.035,$$

and number of payments

$$n = 10.$$

Let X be the amount the grandmother must deposit today. Then

$$X = 1000a_{\overline{10}|}.$$

Using the present value formula for an annuity-immediate,

$$a_{\overline{10}|} = \frac{1 - v^{10}}{i},$$

where

$$v = \frac{1}{1 + i}.$$

Therefore,

$$X = 1000 \left(\frac{1 - (1.035)^{-10}}{0.035} \right).$$

Finally,

$$X \approx 8316.605323.$$

Hence, the grandmother must deposit approximately

$$\boxed{\$8,316.61}.$$